MTH 310 **Exam 3** Name:____________________

Spring, 2026 Sections 15.3–16.6

Instructions. (100 points) Sections 15.3–16.6. You have approximately 70 minutes. You may use a calculator. Show all work for credit. Several problems ask you to **set up but not evaluate**; for those, *full credit will be given only if your setup uses a coordinate system that makes the integral easy to evaluate.* Setups that are correct but unnecessarily difficult will not earn full credit.

(12pts) **1.** Evaluate $\iint_D x\,y\;dA$, where D is the portion of the disk $x^2+y^2 \le 4$ lying in the first quadrant.

Solution:

In polar, $x = r\cos\theta$, $y = r\sin\theta$, $dA = r\,dr\,d\theta$. The first quadrant of the disk is $0 \le r \le 2$, $0 \le \theta \le \pi/2$.

$$\iint_D xy\,dA = \int_0^{\pi/2}\int_0^2 (r\cos\theta)(r\sin\theta)\cdot r\,dr\,d\theta = \int_0^{\pi/2} \cos\theta\sin\theta\,d\theta \cdot \int_0^2 r^3\,dr.$$

$$= \tfrac{1}{2}\cdot 4 = \boxed{2}.$$

(12pts) **2. Set up, but do not evaluate**, an integral for the surface area of the part of the upper hemisphere $z = \sqrt{4-x^2-y^2}$ that lies inside the cylinder $x^2+y^2 = 1$. *Full credit requires a setup whose integrals can be evaluated easily.*

Solution:

Either of the two setups below earns full credit; both lead to the same value $4\pi(2-\sqrt{3})$.

Method 1 (graph, Section 15.5). With $g(x,y) = \sqrt{4-x^2-y^2}$:

$$g_x = \frac{-x}{\sqrt{4-x^2-y^2}}, \quad g_y = \frac{-y}{\sqrt{4-x^2-y^2}}, \quad \sqrt{g_x^2+g_y^2+1} = \frac{2}{\sqrt{4-x^2-y^2}}.$$

The cylinder cuts out the disk $x^2+y^2 \le 1$. In polar:

$$A = \int_0^{2\pi}\int_0^1 \frac{2r}{\sqrt{4-r^2}}\,dr\,d\theta = 2\pi\cdot 2\big[-\sqrt{4-r^2}\big]_0^1 = 4\pi(2-\sqrt{3}).$$

Method 2 (spherical parametrization, Section 16.6). Use $\mathbf{r}(\theta,\phi) = \langle 2\sin\phi\cos\theta,\, 2\sin\phi\sin\theta,\, 2\cos\phi\rangle$ with $|\mathbf{r}_\phi\times\mathbf{r}_\theta| = 4\sin\phi$. The constraint $x^2+y^2\le 1$ becomes $4\sin^2\phi \le 1$, i.e. $\sin\phi \le \frac{1}{2}$, so $0 \le \phi \le \pi/6$.

$$A = \int_0^{2\pi}\int_0^{\pi/6} 4\sin\phi\,d\phi\,d\theta = 8\pi\big(1-\tfrac{\sqrt{3}}{2}\big) = 4\pi(2-\sqrt{3}).$$

$$\boxed{A = 4\pi(2-\sqrt{3}).}$$

24 pts

(12$^{\text{pts}}$) **3. Set up, but do not evaluate**, an integral for the volume of the solid bounded by the paraboloids $y = x^2 + z^2$ and $y = 4 - x^2 - z^2$. *Full credit requires a setup whose integral can be evaluated easily.*

Solution:

Both paraboloids have axis along the y-axis, so use cylindrical coordinates with polar in the xz-plane: $x = r\cos\theta$, $z = r\sin\theta$, $dV = r\,dy\,dr\,d\theta$.

Find the intersection. $r^2 = 4 - r^2 \Rightarrow r = \sqrt{2}$.

Set up. For fixed (r, θ) with $0 \le r \le \sqrt{2}$, y runs from r^2 (forward-opening paraboloid) to $4 - r^2$ (backward-opening paraboloid):

$$\boxed{V = \int_0^{2\pi}\int_0^{\sqrt{2}}\int_{r^2}^{4-r^2} r\,dy\,dr\,d\theta.}$$

(For reference, this evaluates to 4π.)

(12$^{\text{pts}}$) **4. Set up, but do not evaluate**, an integral for the volume of the solid bounded above by the sphere $x^2 + y^2 + z^2 = 4$ and below by the cone $z = \sqrt{3(x^2 + y^2)}$. *Full credit requires a setup whose integral can be evaluated easily.*

Solution:

Identify the cone angle. $z = \sqrt{3}\,r$ corresponds to $\tan\phi = r/z = 1/\sqrt{3}$, so $\phi = \pi/6$. "Inside the cone" means $0 \le \phi \le \pi/6$; the sphere gives $0 \le \rho \le 2$.

Set up (spherical, $dV = \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$):

$$\boxed{V = \int_0^{2\pi}\int_0^{\pi/6}\int_0^{2} \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta.}$$

(For reference, this evaluates to $\frac{16\pi}{3}\left(1 - \frac{\sqrt{3}}{2}\right)$.)

24 pts

(12$^{\text{pts}}$) **5.** Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ where $\mathbf{F}(x,y) = \langle y,\ x+y\rangle$ and C is the parabolic arc $\mathbf{r}(t) = \langle t,\, t^2\rangle$, $0 \le t \le 1$.

Solution:

$\mathbf{r}'(t) = \langle 1,\, 2t\rangle$ and $\mathbf{F}(\mathbf{r}(t)) = \langle t^2,\, t+t^2\rangle$.

$$\mathbf{F}\cdot\mathbf{r}'(t) = t^2\cdot 1 + (t+t^2)\cdot 2t = t^2 + 2t^2 + 2t^3 = 3t^2 + 2t^3.$$

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_0^1 (3t^2+2t^3)\,dt = 1 + \tfrac{1}{2} = \boxed{\tfrac{3}{2}}.$$

(14$^{\text{pts}}$) **6.** Let $\mathbf{F}(x,y) = \langle 2xy + y,\ x^2 + x\rangle$, and let C be the closed path from $(0,0)$ to $(1,1)$ along $y = x^2$, then back from $(1,1)$ to $(0,0)$ along $y = x$.

(a) *(4 pts)* Is $\mathbf{F}$ conservative? If so, what does that imply about $\oint_C \mathbf{F}\cdot d\mathbf{r}$ before doing any computation?

Solution:

$P = 2xy + y$, $Q = x^2 + x$. Then $P_y = 2x+1$ and $Q_x = 2x+1$. Since $P_y = Q_x$ on $\mathbb{R}^2$ (simply connected), $\mathbf{F}$ **is** conservative. Therefore $\oint_C \mathbf{F}\cdot d\mathbf{r} = 0$ for every closed curve, so $\boxed{\oint_C \mathbf{F}\cdot d\mathbf{r} = 0}$.

(b) *(6 pts)* Verify your answer in (a) by applying Green's Theorem.

Solution:

By Green's Theorem,

$$\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_D (Q_x - P_y)\,dA = \iint_D \big((2x+1) - (2x+1)\big)\,dA = \iint_D 0\,dA = \boxed{0}.\ \checkmark$$

The integrand is identically zero because $\mathbf{F}$ is conservative, so the value is 0 regardless of which region D is enclosed.

(c) *(4 pts)* Find a potential function $f(x,y)$ with $\nabla f = \mathbf{F}$.

Solution:

From $f_x = 2xy + y$: $f(x,y) = x^2y + xy + g(y)$. Match $f_y = x^2 + x + g'(y) \overset{!}{=} x^2 + x$, so $g'(y) = 0$.

$$\boxed{f(x,y) = x^2y + xy.}$$

26 pts

(14pts) **7.** Use Green's Theorem with $\mathbf{F}(x,y) = \left\langle -\frac{1}{2}\,y,\ \frac{1}{2}\,x \right\rangle$ to prove that the area enclosed by the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ is πab.

(a) *(6 pts)* Compute $Q_x - P_y$ for this $\mathbf{F}$. What does Green's Theorem then tell you about the value of $\displaystyle\oint_C \mathbf{F} \cdot d\mathbf{r}$ when C is any simple closed curve in the plane oriented counterclockwise?

Solution:

With $P = -\frac{1}{2}y$ and $Q = \frac{1}{2}x$:

$$Q_x - P_y = \tfrac{1}{2} - \left(-\tfrac{1}{2}\right) = 1.$$

By Green's Theorem, with D the region enclosed by C,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (Q_x - P_y)\, dA = \iint_D 1\, dA = \text{Area}(D).$$

So $\oint_C \mathbf{F} \cdot d\mathbf{r}$ equals the area enclosed by C for any simple closed CCW curve.

(b) *(8 pts)* Parameterize the ellipse and compute $\displaystyle\oint_C \mathbf{F} \cdot d\mathbf{r}$ directly. Use part (a) to conclude that the area enclosed by the ellipse is πab.

Solution:

Parametrize. $\mathbf{r}(t) = \langle a\cos t,\, b\sin t\rangle$, $0 \le t \le 2\pi$.
Then $\mathbf{r}'(t) = \langle -a\sin t,\, b\cos t\rangle$ and $\mathbf{F}(\mathbf{r}(t)) = \langle -\frac{b}{2}\sin t,\ \frac{a}{2}\cos t\rangle$.

Dot product.

$$\mathbf{F} \cdot \mathbf{r}'(t) = \left(-\tfrac{b}{2}\sin t\right)(-a\sin t) + \left(\tfrac{a}{2}\cos t\right)(b\cos t) = \tfrac{ab}{2}(\sin^2 t + \cos^2 t) = \tfrac{ab}{2}.$$

Integrate.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \tfrac{ab}{2}\, dt = \pi ab.$$

By part (a), this equals the area enclosed by the ellipse, so $\boxed{\text{Area} = \pi ab}$. $\square$

14 pts

(12pts) **8.** For each statement, decide whether it is **true** or **false**, and give one sentence of justification.

(a) *(3 pts)* If $\mathbf{F}$ is continuously differentiable on $\mathbb{R}^2$ and $P_y = Q_x$ at every point, then $\mathbf{F}$ is conservative.

Solution:

True. $\mathbb{R}^2$ is simply connected, so the cross-partial test is sufficient for conservativeness.

(b) *(3 pts)* If $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for one particular simple closed curve C, then $\mathbf{F}$ is conservative.

Solution:

False. Conservativeness requires zero circulation on *every* closed curve; a single closed loop giving 0 doesn't imply this.

(c) *(3 pts)* If $\operatorname{div} \mathbf{F} = 0$ at every point of a region, then $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve C in that region.

Solution:

False. Vanishing circulation comes from $\operatorname{curl} \mathbf{F} = \mathbf{0}$, not from $\operatorname{div} \mathbf{F} = 0$. Divergence-free fields can still circulate.

(d) *(3 pts)* Green's Theorem applies whenever C is a simple closed curve in the plane and P, Q are continuously differentiable on the region D enclosed by C.

Solution:

True. A simple closed curve in the plane bounds a simply connected region, so the standard hypotheses are satisfied as long as P, Q are smooth on D.

12 pts